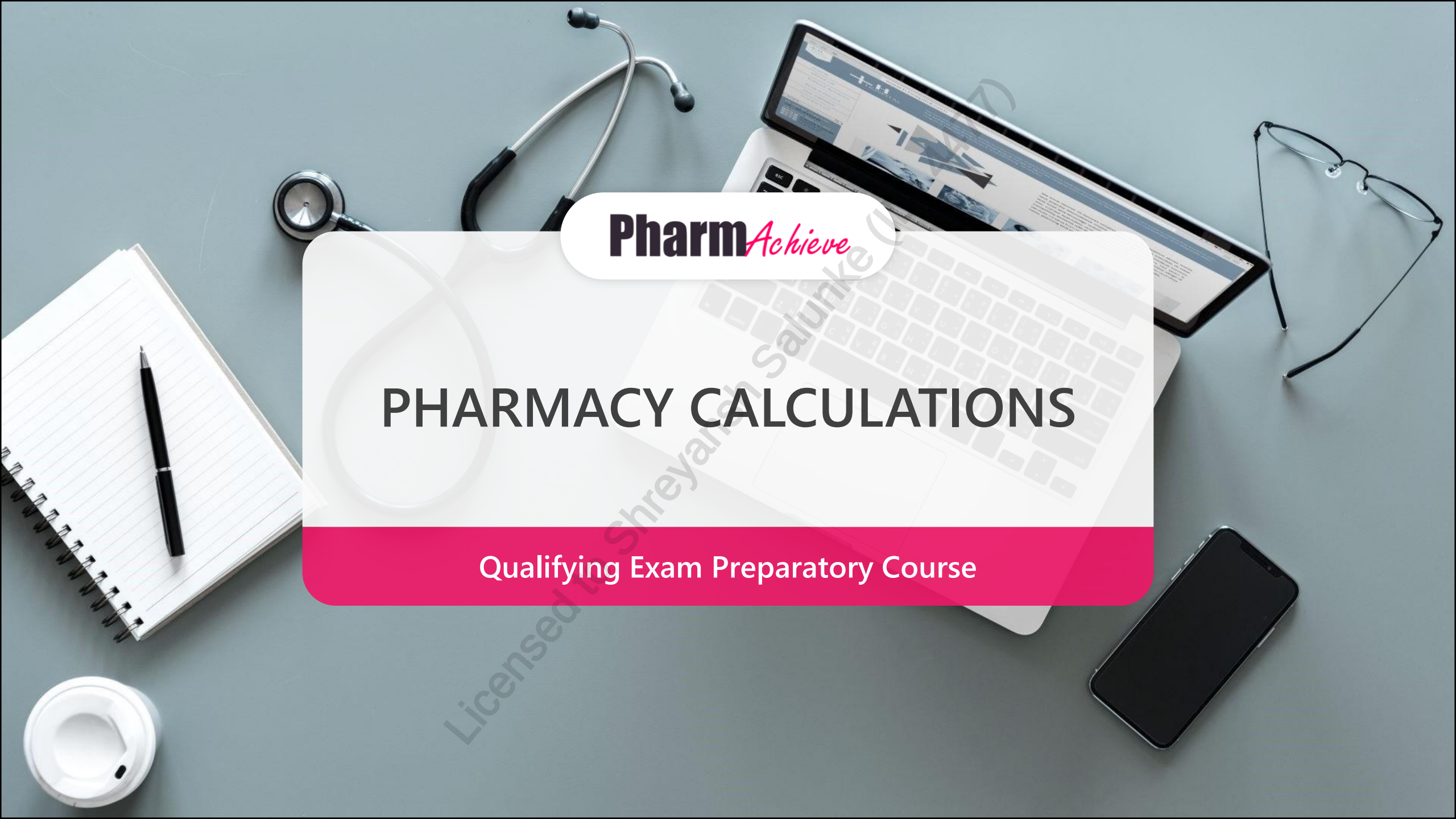




Pharm*Achieve*

PHARMACY CALCULATIONS

Qualifying Exam Preparatory Course

COVERED IN THIS PRESENTATION...

Providing Care:
Clinical Care

Providing Care:
Distribution

Knowledge
& Expertise

Communication
& Collaboration

Leadership &
Stewardship

Professionalism

SUMMARY

Measurement System	Compounding Calculations	Chemistry	Concentration Expressions
Metric System	Solid & Liquid Units in compounding	Mole & Millimole	Mass/Volume
Apothecary system	Density & Specific Gravity	Equivalent weight & Milliequivalent weight	Ratio Strength
Household system	Stock Solutions (& Aliquoting) – Balance sensitivity	Milliosmole	% Strength (W/W; W/V; V/V)
IU	Dilution using diluents	Buffer systems	PPM & PPB
	Dilution without diluents (Alligation)	Isotonicity	Molar concentration
	Fortification	Salt factor	Molal concentration
	Enlarging & Reducing		Normal concentration
	Displacement volume & Density Factor		
	Capsule filling (IR, SR)		

SUMMARY

Dose Calculations	Hospital Pharmacy
$(\text{Dose/unit weight} * \text{weight}) * \text{Frequency} * \text{Treatment Period} = \text{Qty}$	BSA
$\text{Qty} * (\text{Refill number} + 1) = \text{Auth Qty}$	Ideal Body weight
Treatment period	Opioid Conversions
Drops	Infusion Rate (R)
Buffers	TPN
	Parenterals
	Therapeutic Monitoring (CrCl)
	Pediatric Dose

UNITS OF MEASUREMENT

Qualifying Exam Preparatory Course



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UNITS OF MEASURE

SI Units: The international system of units, formally known as the metric system. Contains prefixes to distinguish different sizes for a specific base unit of a certain measurement

Units to measure

- Length (meter: m)
- Volume (liter: L)
- Mass (kilogram: kg)

Each unit of measurement has a base unit. The prefixes listed are used for measurements much smaller or larger than the base unit.

Multiples

- Kilo – 10^3 times the base unit
- Mega – 10^6 times the base unit
- Giga – 10^9 times the base unit
- Tera – 10^{12} times the base unit

Subdivisions

- *Deci – 10^{-1} of the base unit*
- *Centi – 10^{-2} of the base unit*
- Milli – 10^{-3} of the base unit
- Micro – 10^{-6} of the base unit
- Nano – 10^{-9} of the base unit

UNITS OF MEASURE

SI PREFIX		SI SYMBOL
yotta		Y
zetta		Z
exa		E
peta		P
tera		T
giga		G
mega		M
kilo		k
hecto		h
deca		da
deci		d
centi		c
milli		m
micro		μ
nano		n
pico		p
femto		f
atto		a
zepto		z
yocto		y

SI UNIT CONVERSION FACTOR (STANDARD FORM)		FACTOR (POWER)	FACTOR LANGUAGE
1 yottametre	= 1 000 000 000 000 000 000 000 metres	10 ²⁴	septillion
1 zettametre	= 1 000 000 000 000 000 000 000 metres	10 ²¹	sextillion
1 exametre	= 1 000 000 000 000 000 000 metres	10 ¹⁸	quintillion
1 petametre	= 1 000 000 000 000 000 metres	10 ¹⁵	quadrillion
1 terametre	= 1 000 000 000 000 metres	10 ¹²	trillion
1 gigametre	= 1 000 000 000 metres	10 ⁹	billion
1 megametre	= 1 000 000 metres	10 ⁶	million
1 kilometre	= 1 000 metres	10 ³	thousand
1 hectometre	= 100 metres	10 ²	hundred
1 decametre	= 10 metres	10 ¹	ten
1 metre	= 1 metre	10 ⁰	one
1 decimetre	= 0.1 metres	10 ⁻¹	tenth
1 centimetre	= 0.01 metres	10 ⁻²	hundredth
1 millimetre	= 0.001 metres	10 ⁻³	thousandth
1 micrometre	= 0.000 001 metres	10 ⁻⁶	millionth
1 nanometre	= 0.000 000 001 metres	10 ⁻⁹	billionth
1 picometre	= 0.000 000 000 001 metres	10 ⁻¹²	trillionth
1 femtometre	= 0.000 000 000 000 001 metres	10 ⁻¹⁵	quadrillionth
1 attometre	= 0.000 000 000 000 000 001 metres	10 ⁻¹⁸	quintillionth
1 zeptometre	= 0.000 000 000 000 000 000 001 metres	10 ⁻²¹	sextillionth
1 yoctometre	= 0.000 000 000 000 000 000 000 001 metres	10 ⁻²⁴	septillionth

UNITS OF MEASURE

INTERSYSTEM CONVERSIONS

- The meter is the primary unit in the SI system.
- One liter is the volume of a cube one tenth of a meter:

1 litre = 1 dm³

1 mL = 1 cm³

- 1g = 1mL of water at 4°C



Note: drachm (ʒ; ≈4) and ounce (ʒ;≈30) can be used for weight (g) and volume (ml) measurements.

Common SI and Imperial Conversions

Equivalents of Length

1 inch	2.54cm
1 Ft	12 in
1 metre	39.37 in

Equivalents of Volume

1 teaspoon (tsp.)	5 mL
1 tablespoon (tbsp.)	15 mL
1 fluidounce (fl. oz.)	29.57 mL
1 pint (16 fl. oz.)	473 mL
1 quart (32 fl. oz.)	946 mL
1 gallon, US (128 fl. oz.)	3785 mL

Equivalents of Weight

1 pound (lb)	454 g
1 kilogram	2.2 lb

DOSAGE CALCULATIONS

Qualifying Exam Preparatory Course



Pharm*Achieve*

DOSAGE CALCULATIONS

Terms:

- **Median effective dose (ED₅₀)** = amount that produces desired effect in 50% of subjects.
- **Median toxic dose (TD₅₀)** = amount that produces toxic effects in 50% of subjects
- **Minimum effective concentration (MEC)** = minimum blood serum concentration expected to give the desired effect.
- **Maximum tolerated dose (MTC)** = maximum blood serum concentration before undesired effects occur.
- **Therapeutic window (MTC-MEC)** = the serum concentration where a medication is effective but not toxic. It is the concentration between the MTC and MEC.

DOSAGE CALCULATIONS

* Dosing based on body weight and surface area

$$\text{Patient's dose (mg)} = \text{Patient's weight (kg)} * \frac{\text{Drug dose (mg)}}{1 \text{ kg}}$$

Example

The usual initial dose of drug X is 150µg/kg. How many milligrams should be administered to a 154 lb patient?

Example

JC is a 9-year-old boy who is prescribed a chemotherapeutic agent. His body surface area (BSA) is 0.87m². The usual adult dose for this drug is 240 mg/day. Assuming the chemotherapeutic agent is dose-adjusted according to BSA and normal adult BSA is 1.73m², what should JC's daily dose be?

DOSAGE CALCULATIONS

* Dosing based on body weight and surface area

$$\text{Patient's dose (mg)} = \text{Patient's weight (kg)} * \frac{\text{Drug dose (mg)}}{1 \text{ kg}}$$

Example

The usual initial dose of drug X is 150µg/kg. How many milligrams should be administered to a 154 lb patient?

$$150\mu\text{g} = 0.15\text{mg}$$

$$1\text{kg} = 2.2\text{lb}$$

$$\text{Patient's dose} = 154\text{lb} * \frac{0.15\text{mg}}{2.2\text{lb}}$$

$$\text{Patient's dose} = 10.5 \text{ mg}$$

Example

JC is a 9-year-old boy who is prescribed a chemotherapeutic agent. His body surface area (BSA) is 0.87m². The usual adult dose for this drug is 240 mg/day. Assuming the chemotherapeutic agent is dose-adjusted according to BSA and normal adult BSA is 1.73m², what should JC's daily dose be?

$$\begin{aligned}\text{JC's dose} &= \frac{\text{JC's BSA}}{\text{Normalized Adult BSA}} \times \text{usual adult dose} \\ &= \frac{0.87\text{m}^2}{1.73\text{m}^2} \times 240 \frac{\text{mg}}{\text{day}} \\ &= 120.7 \frac{\text{mg}}{\text{day}}\end{aligned}$$

DOSAGE CALCULATIONS

* Calculating required volume or dose of a given concentration of a suspension

Example 1(a)

AC, a 6-year-old girl, weighs 60 lbs. She was diagnosed with otitis media and prescribed amoxicillin 50 mg/kg body weight/day x 7 days. How many bottles of amoxicillin suspension 50 mg/mL (each bottle containing 200 mL upon reconstitution) need to be dispensed for this prescription?

Example 1(b)

AC is prescribed amoxicillin three times per day (TID). How many mL per dose should AC be given?

DOSAGE CALCULATIONS

* Calculating required volume or dose of a given concentration of a suspension

Example 1(a)

AC, a 6-year-old girl, weighs 60 lbs. She was diagnosed with otitis media and prescribed amoxicillin 50 mg/kg body weight/day x 7 days. How many bottles of amoxicillin suspension 50 mg/mL (each bottle containing 200 mL upon reconstitution) need to be dispensed for this prescription?

1. AC's mass in kg:

$$\frac{60 \text{ lbs}}{2.2 \frac{\text{lbs}}{\text{kg}}} = 27.3 \text{ kg}$$

2. Amoxicillin needed for regimen:

$$\frac{50 \text{ mg}}{\text{kg} \cdot \text{day}} \times 7 \text{ days} \times 27.3 \text{ kg} = 9555 \text{ mg}$$

3. Volume needed:

$$\frac{9555 \text{ mg}}{50 \frac{\text{mg}}{\text{mL}}} = 191.1 \text{ mL}$$

4. Bottles needed:

$$\frac{191.1 \text{ mL}}{200 \text{ mL}} = 1 \text{ bottle}$$

Example 1(b)

AC is prescribed amoxicillin three times per day (TID). How many mL per dose should AC be given?

1. AC's daily dose:

$$\frac{191.1 \text{ mL}}{7 \text{ days}} = \frac{X}{\text{day}} \quad X = 27.3 \text{ mL}$$

2. Dose divided into 3 daily doses:

$$\frac{27.3 \text{ mL/day}}{3 \text{ doses/day}} = 9.1 \text{ mL/dose}$$

DOSAGE CALCULATIONS

- * Calculating the dispensed amount of medication when exact quantities may not be possible

Example

Patient OP comes into the pharmacy looking for cromolyn sodium eye drops to relieve her red eyes. The doctor told her to instill 1 gtt QID ou. OP is planning to go out of town for 30 days. If the bottle of cromolyn sodium eye drops has 5mL, how many bottles does OP need to last her for 30 days? Assume there are 15 drops of cromolyn sodium per mL of solution.

DOSAGE CALCULATIONS

* Calculating the dispensed amount of medication when exact quantities may not be possible

Example

Patient OP comes into the pharmacy looking for cromolyn sodium eye drops to relieve her red eyes. The doctor told her to instill 1 gtt QID ou. OP is planning to go out of town for 30 days. If the bottle of cromolyn sodium eye drops has 5mL, how many bottles does OP need to last her for 30 days? Assume there are 15 drops of cromolyn sodium per mL of solution.

1. 1 bottle = $5 \text{ mL} \times 15 \frac{\text{drops}}{\text{mL}} = 75 \text{ drops}$

2. # of daily drops = $1 \text{ drop} \times 2 \text{ eyes} \times 4 \frac{\text{times}}{\text{day}} = 8 \frac{\text{drops}}{\text{day}}$

3. Cross multiply $\frac{1 \text{ day}}{8 \text{ drops}} = \frac{30 \text{ days}}{x} \quad x = 240 \text{ drops}$

4. $\frac{240 \text{ drops}}{75 \frac{\text{drops}}{\text{bottle}}} = 3.2 \text{ bottles} = 4 \text{ bottles}$

DOSAGE CALCULATIONS

- * Calculating the dispensed amount of medication when exact quantities may not be possible

Example

Mrs. CD comes into the pharmacy with a prescription for Symbicort® (formoterol 6mcg and budesonide 100mcg) 1 puff BID. If there are 120 doses in a turbuhaler, how many turbuhalers do you need to dispense for a 3-month supply for Mrs. CD?

DOSAGE CALCULATIONS

* Calculating the dispensed amount of medication when exact quantities may not be possible

Example

Mrs. CD comes into the pharmacy with a prescription for Symbicort® (formoterol 6mcg and budesonide 100mcg) 1 puff BID. If there are 120 doses in a turbuhaler, how many turbuhalers do you need to dispense for a 3-month supply for Mrs. CD?

$$\frac{2 \text{ puffs}}{\text{day}} \times 90 \text{ days} = 180 \text{ puffs}$$

$$\frac{180 \text{ puffs}}{120 \frac{\text{puffs}}{\text{turbuhaler}}} = 1.5 = 2 \text{ turbuhalers}$$

DOSAGE CALCULATIONS

OPIOID CONVERSIONS

- Step 1: Ensure the formulation is consistent (PO vs. IV)
- Step 2: Use the chart to convert from one opioid to another
- Step 3: Reduce the new dose by 25-50% due to incomplete cross-tolerance (few exceptions)

Drug	Oral dose (mg)
Morphine	30
Oxycodone	20
Hydromorphone	6
Codeine	200
Tramadol and Methadone	Morphine dose equivalence not reliably established

DOSAGE CALCULATIONS

OPIOID CONVERSIONS

Example

Convert 160 mg/day of morphine to its oxycodone dose equivalence.

DOSAGE CALCULATIONS

OPIOID CONVERSIONS

Example

Convert 160 mg/day of morphine to its oxycodone dose equivalence.

$$\frac{30 \text{ mg morphine}}{20 \text{ mg oxycodone}} = \frac{160 \text{ mg morphine}}{x \text{ mg oxycodone}}$$

$$x \text{ mg oxycodone} = \frac{160 \text{ mg morphine} \times 20 \text{ mg oxycodone}}{30 \text{ mg morphine}}$$

$$\begin{aligned} x \text{ mg oxycodone} &= 106.67 \text{ mg} \times 50\% = 53.34 \text{ mg} \\ x \text{ mg oxycodone} &= 106.67 \text{ mg} \times 25\% = 26.7 \text{ mg} \\ x \text{ mg oxycodone} &= 106.67 \text{ mg} - 26.7 \text{ mg} \\ &= 79.97 \text{ mg} \end{aligned}$$

DOSAGE CALCULATIONS

FENTANYL CONVERSIONS

Morphine PO (mg per day)	Fentanyl patch (mcg per hour)
60-134	25
135-179	37
180-224	50
225-269	62
270-314	75
315-359	87
360-404	100
405-494	125
495-584	150
585-674	175
675-764	200

Morphine PO (mg per day)	Fentanyl patch (mcg per hour)
765-854	225
855-944	250
945-1034	275
1035-1124	300

Note: When converting from morphine to fentanyl, there is **no need** for a 25%-50% dose reduction since incomplete cross-tolerance is already accounted for in the conversion chart.

REVIEW QUESTION #1 – OPIOID CONVERSION

AT, a 45-year-old male, has been receiving hydromorphone 1.5 mg IV every 4 hours for severe post-operative pain. As his pain is improving, he is being prepared for discharge and the physician wants to switch him to oral morphine for continued pain management at home. Options for oral morphine include extended-release formulation every 12 hours or immediate-release formulation every 4-6 hours.

- I. What is the total daily dose of IV Hydromorphone that AT is currently receiving?
 - a) 6 mg/day
 - b) 9 mg/day
 - c) 36 mg/day
 - d) 12 mg/day

- II. What is the equivalent daily dose of oral morphine that AT would need to be taking?
 - a) 180 mg/day
 - b) 145 mg/day
 - c) 135 mg/day
 - d) 170 mg/day

REVIEW QUESTIONS – OPIOID CONVERSION

AT, a 45-year-old male, has been receiving hydromorphone 1.5 mg IV every 4 hours for severe post-operative pain. As his pain is improving, he is being prepared for discharge and the physician wants to switch him to oral morphine for continued pain management at home. Options for oral morphine include extended-release formulation every 12 hours or immediate-release formulation every 4-6 hours.

II. What is the equivalent daily dose of oral morphine that AT would need to be taking?

- a) 180 mg/day
- b) 145 mg/day
- c) 135 mg/day
- d) 170 mg/day

Drug	Oral dose (mg)	IV/IM/SC dose (mg)
Morphine	30	10
Oxycodone	20	-
Hydromorphone	6	1.5
Codeine	200	-

REVIEW QUESTIONS – OPIOID CONVERSION

AT, a 45-year-old male, has been receiving hydromorphone 1.5 mg IV every 4 hours for severe post-operative pain. As his pain is improving, he is being prepared for discharge and the physician wants to switch him to oral morphine for continued pain management at home. Options for oral morphine include extended-release formulation every 12 hours or immediate-release formulation every 4-6 hours.

III. What would be an appropriate oral morphine dose and regimen? You have 10, 15, 30, 60, 75, 100 or 200 mg extended-release capsules available.

- a) 75 mg qam + 60 mg qpm
- b) 30 mg BID
- c) 70 mg BID
- d) 90 mg BID

REVIEW QUESTIONS – OPIOID CONVERSION

AT, a 45-year-old male, has been receiving hydromorphone 1.5 mg IV every 4 hours for severe post-operative pain. As his pain is improving, he is being prepared for discharge and the physician wants to switch him to oral morphine for continued pain management at home. Options for oral morphine include extended-release formulation every 12 hours or immediate-release formulation every 4-6 hours.

IV. What would be an appropriate dose for fentanyl patches, if the patient is unable to swallow (NPO) upon discharge? Fentanyl is currently available in up to 200mcg/hr doses.

- a) 25 mcg/hr
- b) 37 mcg/hr
- c) 50 mcg/hr
- d) 75 mcg/hr

Morphine PO (mg per day)	Fentanyl patch (mcg per hour)
60-134	25
135-179	37
180-224	50
225-269	62
270-314	75
315-359	87
360-404	100
405-494	125
495-584	150
585-674	175
675-764	200

DOSING FOR SPECIFIC POPULATIONS

Qualifying Exam Preparatory Course



Pharm*Achieve*

BODY MASS INDEX (BMI)

Clinical standard for judging excessive weight and obesity

- Usually expressed without units; however, the understood units are kg/m²
- Assists in determining if a patient is obese and requires weight-based dose adjustments (for lipophilic drugs that distribute into fat tissues)
- Accurately categorizes patients according to their optimal weight range (not really)

$$\text{BMI} = \frac{\text{Weight (kg)}}{[\text{Height (m)}]^2}$$

BMI Range	Weight Status
< 18.5	Underweight
18.5 – 24.9	Normal
25 – 29.9	Overweight
> 30	Obese

BODY MASS INDEX (BMI)

Example 1

Calculate the BMI of a person 6' in height and weighing 200 lb.

Example 2

Calculate the BMI of a person 7' 0" in height and weighing 235 lb

BODY MASS INDEX (BMI)

Example 1

Calculate the BMI of a person 6' in height and weighing 200 lb.

$$\text{BMI} = \frac{\text{Weight (kg)}}{[\text{Height (m)}]^2}$$
$$200 \text{ lb} \times \frac{1 \text{ Kg}}{2.2 \text{ lb}} = 90.9 \text{ Kg}$$
$$6' = 72 \text{ in} \rightarrow 72 \text{ in} \times \frac{2.54 \text{ cm}}{1 \text{ in}} \times \frac{1 \text{ m}}{100 \text{ cm}} = 1.83 \text{ m}$$
$$\text{BMI} = \frac{90.9 \text{ Kg}}{(1.83 \text{ m})^2} = 27.1 \text{ (overweight)}$$

Example 2

Calculate the BMI of a person 7' 0" in height and weighing 235 lb

$$\text{BMI} = \frac{\text{Weight (kg)}}{[\text{Height (m)}]^2}$$
$$235 \text{ lb} \times \frac{1 \text{ Kg}}{2.2 \text{ lb}} = 106.8 \text{ kg}$$
$$7' 0'' = 84 \text{ in} \rightarrow 84 \text{ in} \times \frac{2.54 \text{ cm}}{1 \text{ in}} \times \frac{1 \text{ m}}{100 \text{ cm}} = 2.13 \text{ m}$$
$$\text{BMI} = \frac{106.8 \text{ Kg}}{(2.13 \text{ m})^2} = 23.5 \text{ (normal weight)}$$

IDEAL BODY WEIGHT

* Ideal body weight formulas are dependent upon the gender and the height of the patient

For women: $45.5\text{kg} + 2.3\text{kg} \times \text{number of inches above 5 feet in height}$

For men: $50\text{kg} + 2.3\text{kg} \times \text{number of inches above 5 feet in height}$

Example

Calculate the IBW of a female who measures 6' in height and weighing 200 lb.

IDEAL BODY WEIGHT

* Ideal body weight formulas are dependent upon the gender and the height of the patient

For women: "45.5kg + 2.3kg x number of inches above 5 feet in height"

For men: "50kg + 2.3kg x number of inches above 5 feet in height"

Example

Calculate the IBW of a female who measures 6' in height and weighing 200 lb.

For women: $45.5\text{kg} + 2.3\text{kg} \times \text{number of inches above 5 feet in height}$

5 feet = 60 inches

6 feet = 72 inches

Number of inches above 5 feet is 12 inches

$$\begin{aligned}\text{IBW} &= 45.5\text{kg} + 2.3\text{kg} \times \text{number of inches above 5 feet in height} \\ &= 45.5\text{kg} + 2.3\text{kg} \times 12 \text{ inches} \\ &= 73.1\text{kg}\end{aligned}$$

This is significantly less than the patient's total weight of 200lb = 90.9 kg

HOW DO WE DECIDE WHICH WEIGHT TO USE?

Calculate IBW:

Males

IBW = 50 kg + 2.3 kg for each inch over 5 feet

Females

IBW = 45.5 kg + 2.3 kg for each inch over 5 feet

Scenario #1: If $ABW < IBW$, use ABW

Scenario #2: If $IBW < ABW < 130\%$ of IBW, use IBW

Scenario #3: If $ABW > 130\%$ of IBW, use AdjBW

AdjBW = $IBW + 0.4 \times (ABW - IBW)$

ABW: Actual Body Weight

AdjBW: Adjusted Body Weight

IBW: Ideal Body Weight

CREATININE CLEARANCE

COCKCROFT-GAULT EQUATION

Cockcroft-Gault estimates a patient's renal function

- Cockcroft-Gault equation estimates renal function by taking into consideration patient's age, gender, ideal body mass, and serum creatinine.
- May overestimate the glomerular filtration rate (GFR) in patients who have rapidly-declining renal function.
- May underestimate GFR in patients who are rapidly recovering from an illness/condition that initially lowered their renal function.

$$\text{CrCl (mL/min)} = \frac{(140 - \text{age}) \times 1.2 \times \text{body weight (kg)}}{\text{serum creatinine} \left(\frac{\mu\text{mol}}{\text{L}} \right)} \quad \times 0.85 \text{ (if female)}$$

$$\text{CrCl (mL/min)} = \frac{(140 - \text{age}) \times \text{body weight (kg)}}{72 \times \text{serum creatinine} \left(\frac{\text{mg}}{\text{dL}} \right)} \quad \times 0.85 \text{ (if female)}$$

CREATININE CLEARANCE

RENAL ADJUSTMENT OF DOSES

Example

Patient AK is a 65-year-old male who weighs 70 kg and measures 5 feet and 10 inches (1.78m). His blood work shows that he has a serum creatinine of 85 $\mu\text{mol/L}$. His doctor wants to prescribe enoxaparin for him, which requires a dose adjustment for CrCl less than 30mL/min. Using the Cockcroft-Gault formula, estimate AK's creatinine clearance and determine whether dose adjustment is needed.

For men: $50\text{kg} + 2.3\text{kg} \times \text{number of inches above 5 feet in height}$

$$\text{CrCl (mL/min)} = \frac{(140 - \text{age}) \times 1.2 \times \text{body weight (kg)}}{\text{serum creatinine } (\mu\text{mol/L})} \times 0.85 \text{ (ONLY if female)}$$

CREATININE CLEARANCE

RENAL ADJUSTMENT OF DOSES

Example

Patient AK is a 65-year-old male who weighs 70 kg and measures 5 feet and 10 inches (1.78m). His blood work shows that he has a serum creatinine of 85 $\mu\text{mol/L}$. His doctor wants to prescribe enoxaparin for him, which requires a dose adjustment for CrCl less than 30mL/min. Using the Cockcroft-Gault formula, estimate AK's creatinine clearance and determine whether dose adjustment is needed.

1. Find IBW

For men: $50\text{kg} + 2.3\text{kg} \times \text{number of inches above 5 feet in height}$

$$50\text{ kg} + (2.3\text{kg} \times 10) = 73\text{kg}$$

2. Solve for CrCl – we use actual body weight since it is less than the ideal body weight

$$\begin{aligned}\text{CrCl}(\text{mL/min}) &= \frac{(140 - \text{age}) \times 1.2 \times \text{body weight (kg)}}{\text{serum creatinine } (\mu\text{mol/L})} = \frac{(140 - 65) \times 1.2 \times (70\text{ kg})}{85 (\mu\text{mol/L})} \\ &= 74.118\text{ mL/min}\end{aligned}$$

Therefore, no, patient AK does not require dose adjustment for enoxaparin based on the current serum creatinine level.

PEDIATRIC DOSE

PEDIATRIC ADJUSTMENT OF DOSES

In practice, these dosing rules are only used if a pediatric dosing reference is not available

Example

HX is an 8-month-old infant who weighs 10kg and needs diphenhydramine. The usual dose of diphenhydramine for an average adult is 25mg PO q4-6h. He asks you to calculate the pediatric dose for HX using Young's rule and Clark's rule because he will prescribe the smaller dose of the two for HX.

Young's Rule (age based) Pediatric dose = dose for average adult $\times \frac{\text{age in years}}{(\text{age in years} + 12)}$

Clark's Rule (mass based) Pediatric dose = dose for average adult $\times \frac{\text{mass of child in lbs}}{150}$

PEDIATRIC DOSE

PEDIATRIC ADJUSTMENT OF DOSES

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Example

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Young's Rule

$$\text{Pediatric dose} = \text{dose for average adult} \times \frac{\text{age in years}}{(\text{age in years} + 12)} = 25 \text{ mg} \times \frac{0.67}{12.67} = 1.32 \text{ mg}$$

Clark's Rule

$$\text{Pediatric dose} = \text{dose for average adult} \times \frac{\text{mass of child in lbs}}{150} = 25 \text{ mg} \times \frac{22}{150} = 3.67 \text{ mg}$$

Therefore, the dose to recommend for HX is 1.32 mg q4-6h.

COMPOUNDING

Qualifying Exam Preparatory Course



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COMPOUNDING

Mixing raw materials with active drug substances to create a drug product.

- Preparation of drug dosage forms.
- Manipulations of commercial products that may require one or more ingredients added.
- Preparation of drugs or devices to the purposes of research, teaching or chemical analysis.

Ratio

- Comparison of part to the whole, like a fraction.
- Ratio strength consists of two numbers: the first being the part and the second the total.
- E.g. 5% in strength represents the ratio of 5 parts in 100 parts, or the ratio 5:100.

Proportion

- The relationship between two ratios.
- Can be thought of as equal fractions when proportioning ratios

Example

If one tablet contains 150 mg of active ingredient and the patient is prescribed 6 tablets a day how much active ingredient will they receive in one day?

150mg :1tablet

$x = 150\text{mg} \times 6$

$x = 900 \text{ mg}$

COMPOUNDING

CALCULATING WEIGHTS USING SPECIFIC GRAVITY

- Specific gravity is a ratio that expresses how many times heavier (or lighter) a liquid is than water. Specific gravity of water is 1
- Density of water is 1g/mL
- Knowing specific gravity of a liquid allows you to convert its mass into volume or vice-versa

$$\text{Specific gravity} = \frac{\text{Weight of substance}}{\text{Weight of equal volume of water}}$$

Example

What is the weight (in grams) of 0.5L of an alcohol with specific gravity of 0.75?

COMPOUNDING

CALCULATING WEIGHTS USING SPECIFIC GRAVITY

- Specific gravity is a ratio that expresses how many times heavier (or lighter) a liquid is than water. Specific gravity of water is 1
- Density of water is 1g/mL
- Knowing specific gravity of a liquid allows you to convert its mass into volume or vice-versa

$$\text{Grams of Alcohol} = \text{Grams of Equal Volume of Water} \times \text{Specific Gravity}$$

Example

What is the weight (in grams) of 0.5L of an alcohol with specific gravity of 0.75?

$$\begin{aligned}\text{Grams of Alcohol} &= \text{Grams of Equal Volume of Water} \times \text{Specific Gravity} \\ &= 500 \text{ g} \times 0.75 \\ &= 375 \text{ g}\end{aligned}$$

COMPOUNDING

- * Calculating quantities based on given ratios, percentages, or concentrations of active ingredients

Strengths of medications are often given in the following formats, and understanding how to convert between these formats and the mass or volume is a useful skill set:

- % volume per volume (v/v) is expressed as mL per 100 mL of solution
- % weight per volume (w/v) is expressed as grams per 100 mL of solution
- % weight per weight (w/w) is expressed as grams of solute per 100 grams of preparation

Example A

A suspension of drug C is 23% w/v. What is the concentration of this suspension in g/L?

Example B

How many grams of NaCl is needed to make 2L of a 0.9% w/v solution?

COMPOUNDING

* Calculating quantities based on given ratios, percentages, or concentrations of active ingredients

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- % weight per volume (w/v) is expressed as grams per 100 mL of solution
- % weight per weight (w/w) is expressed as grams of solute per 100 grams of preparation

Example A

A suspension of drug C is 23% w/v. What is the concentration of this suspension in g/L?

1. Weight/volume

23% w/v means $\frac{23 \text{ g}}{100 \text{ mL}}$

2. Cross-multiplication

$$\frac{23 \text{ g}}{100 \text{ mL}} = \frac{x}{1000 \text{ mL}} \quad x = 230\text{g} \rightarrow 230\text{g/L}$$

Example B

How many grams of NaCl is needed to make 2L of a 0.9% w/v solution?

0.9% w/v means 0.9 g of NaCl /100mL

$$\frac{0.9 \text{ g}}{100 \text{ mL}} \times 2000 \text{ mL} = 18 \text{ g}$$

COMPOUNDING

DILUTION

Concentration

- A measure of the amount of dissolved substance contained per unit volume of a solution
- Common measures:
 - Percent: weight or volume
 - Ratio: 4mL : 30mL
 - Molarity: Mols per liter
 - **Parts per billion (ppb) and parts per million (ppm)**
 - Used for very **dilute solutions**
 - Expressed as both weight/weight (w/w) & weight/volume (w/v)
 - $1 \text{ ppm} = 1 \text{ mg/kg} = 1 \text{ mg/L} = 1 \mu\text{g/g} = 1 \mu\text{g/mL}$
 - $1 \text{ ppb} = 1 \mu\text{g/kg} = 1 \mu\text{g/L} = 1 \text{ ng/g} = 1 \text{ ng/mL}$

Example

Example: 150 mL of an aqueous NaCl solution contains 0.0045 g NaCl. What is the concentration of NaCl in ppm?

Example

A solution of KCl has a concentration of $0.5 \mu\text{g/mL}$ what is the concentration of KCl in ppb?

COMPOUNDING

DILUTION

Concentration

- A measure of the amount of dissolved substance contained per unit volume of a solution
- Common measures:
 - Percent: weight or volume
 - Ratio: 4mL : 30mL
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 - $1 \text{ ppb} = 1 \mu\text{g/kg} = 1 \text{ ng/g} = 1 \mu\text{g/L} = 1 \text{ ng/mL}$

Example

Example: 150 mL of an aqueous NaCl solution contains 0.0045 g NaCl. What is the concentration of NaCl in ppm?

$$0.0045 \text{ g} = 4.5 \text{ mg}$$

$$150 \text{ mL} = 0.15 \text{ L}$$

$$\text{ppm (w/v)} = \frac{\text{mg}}{\text{L}} = \frac{4.5 \text{ mg}}{0.15 \text{ L}} = 30 \text{ ppm}$$

Example

A solution of KCl has a concentration of $0.5 \mu\text{g/mL}$ what is the concentration of KCl in ppb?

$$0.5 \mu\text{g} = 500 \text{ ng}$$

$$\text{ppb (w/v)} = 1 \text{ ng/mL} = 500 \text{ ppb}$$

COMPOUNDING

DILUTION

- Reducing the strength of a pharmaceutical preparation
- Most often done with solutions, but can also be done with powders or ointments

$$C_i V_i = C_f V_f$$

C=concentration
V=volume
i=initial
f=final

Example

If 500mL of a 15% v/v solution are diluted to 1500mL what will be the final percent strength v/v?

Example

How many milliliters of a 1:400 w/v stock solution should be used to make 4 liters of a 1:2000 w/v solution?

COMPOUNDING

DILUTION

- Reducing the strength of a pharmaceutical preparation
- Most often done with solutions, but can also be done with powders or ointments
- $V_f - V_i = \text{diluent}$

$$C_i V_i = C_f V_f$$

C=concentration
V=volume
i=initial
f=final

Example

If 500mL of a 15% v/v solution are diluted to 1500mL what will be the final percent strength v/v?

$$C_i V_i = C_f V_f$$

$$(15\%)(500 \text{ mL}) = C_f (1500 \text{ mL})$$

$$C_f = \frac{15\% \times 500 \text{ mL}}{1500 \text{ mL}}$$

$$C_f = 5\%$$

Example

How many milliliters of a 1:400 w/v stock solution should be used to make 4 liters of a 1:2000 w/v solution?

$$C_i V_i = C_f V_f$$

$$\left(\frac{1}{400}\right) V_i = \left(\frac{1}{2000}\right) (4000 \text{ mL})$$

$$V_i = \frac{\frac{1}{2000} \times 400 \text{ mL}}{\frac{1}{400}}$$

$$V_i = 800 \text{ mL}$$

COMPOUNDING

Alligation: An arithmetical method of solving problems involving the mixing solutions with different percent strengths

Alligation Medial

- Calculated by a weighted average of the constituents
- Must know the concentration and quantity of each solution added to the total

Example

What is the percentage (v/v) of alcohol in a mixture of 3000mL of 40%, 1000mL of 60% and 1000mL of 70%

$$0.4 \times 3000 \text{ mL} = 1200 \text{ mL}$$

$$0.6 \times 1000 \text{ mL} = 600 \text{ mL}$$

$$0.7 \times 1000 \text{ mL} = 700 \text{ mL}$$

Totals:	5000 mL	2500 mL
	Total Volume	Total Alcohol Volume

$$\frac{2500 \text{ mL}}{5000 \text{ mL}} = 0.5 = 50\% \text{ (v/v)}$$

Alligation Alternate

- Method used to calculate the number of parts of two or more components to prepare a mixture of a given strength

Example

How many millilitres of 50% alcohol and how many millilitres of 5% alcohol are needed to make a 4500mL solution of 10% alcohol?

$$x + y = 4500 \text{ mL} \text{ (1)}$$

$$0.5x + 0.05y = 4500 \text{ mL} \times 0.1 \text{ (2)}$$

Rearrange (1) for y and sub into (2)

$$0.5x + 0.05(4500 \text{ mL} - x) = 450 \text{ mL}$$

$$0.45x = 225 \text{ mL}$$

$$x = 500 \text{ mL}$$

$$y = 4500 \text{ mL} - 500 \text{ mL} = 4000 \text{ mL}$$

COMPOUNDING

ALLIGATION: SYSTEM OF EQUATIONS METHOD

A pharmacist wishes to use three different dextrose solutions to create 500mL of a 10% dextrose solution. The three solutions the pharmacist wishes to use are; 50%, 20% and 5%. How much of each solution should the pharmacist use?

Step 1

There are many combinations for the three solutions

Step 2

Set up system of equations

Step 3

Solve system of equations

COMPOUNDING

ALLIGATION: SYSTEM OF EQUATIONS METHOD

A pharmacist wishes to use three different dextrose solutions to create 500mL of a 10% dextrose solution. The three solutions the pharmacist wishes to use are; 50%, 20% and 5%. How much of each solution should the pharmacist use?

Step 1

There are many combinations for the three solutions, for the easiest calculations assume the amount of 50% and 20% are equal

Step 2

Set up system of equations:

A=50% solution, B=20%, C=5%

$$2A + C = 500 \quad (1)$$

Assume A=B

$$0.7A + 0.05C = 0.1 \times 500 \text{ mL} \quad (2)$$

Step 3

Solve system of equations:

$$0.7A + 0.05(500 - 2A) = 50 \text{ mL}$$

$$C = 500 - 2 \times 41.67$$

$$C = 416.66 \text{ mL}$$

$$B = A = 41.67 \text{ mL}$$

COMPOUNDING

ALLIGATION: CROSS SUBTRACT METHOD

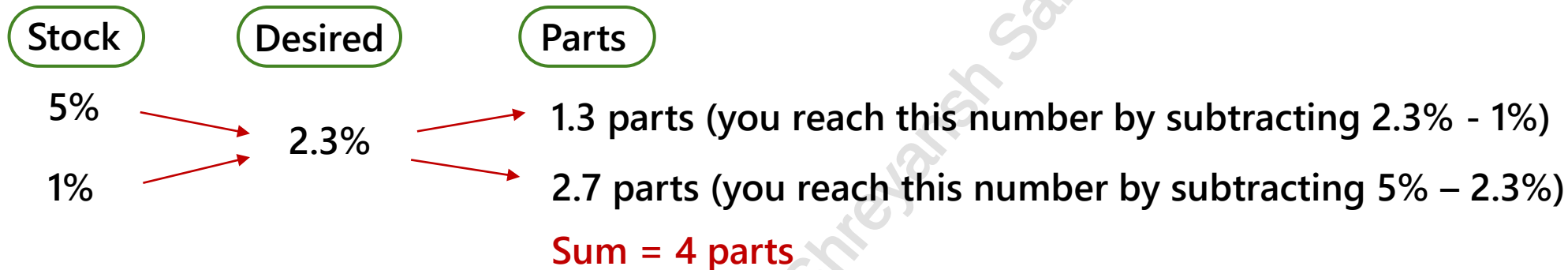
EJ comes in with a prescription for 2.3% hydrocortisone ointment (Mitte: 60g). You have hydrocortisone 1% and 5% stock ointments available. How would you compound this ointment for EJ?

COMPOUNDING

ALLIGATION: CROSS SUBTRACT METHOD

EJ comes in with a prescription for 2.3% hydrocortisone ointment (Mitte: 60g). You have hydrocortisone 1% and 5% stock ointments available. How would you compound this ointment for EJ?

Cross-subtract using the following schematic:



For the 5% ointment:

$$\frac{1.3 \text{ parts}}{4 \text{ total parts}} = 32.5\%$$

$$32.5\% \text{ of } 60\text{g} = 19.5\text{g}$$

For the 1% ointment:

$$\frac{2.7 \text{ parts}}{4 \text{ total parts}} = 67.5\%$$

$$67.5\% \text{ of } 60\text{g} = 40.5\text{g}$$

Therefore, to compound EJ's ointment, you would mix **19.5g of the 5% stock ointment** with **40.5g of the 1% stock ointment** in order to make 60g of 2.3% ointment

COMPOUNDING

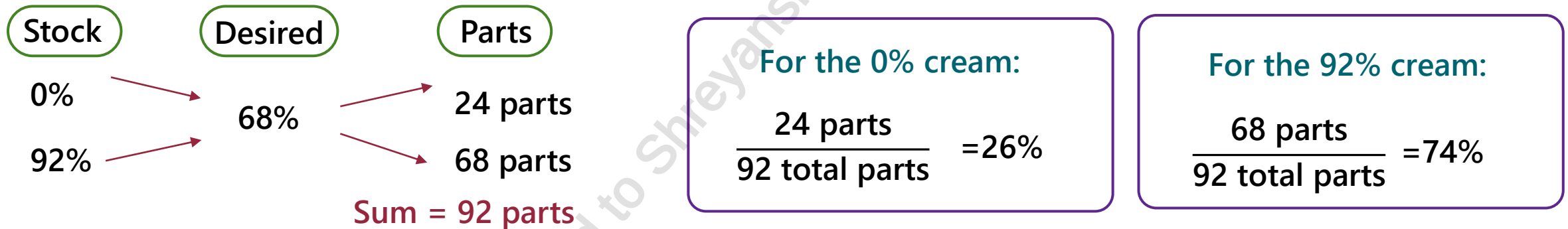
ALLIGATION: SYSTEM OF EQUATIONS METHOD

EJ has an order prescription in the system to be dispensed as needed for athlete's foot. The prescription is for 68% of terbinafine cream in a corticosteroid base. You have 92% terbinafine cream as stock formulation, with only 14g left, as the product seems to be on back-order. You have more than enough corticosteroid base in stock. What is the maximum quantity (mass) of 68% terbinafine cream that you can compound for EJ right now?

COMPOUNDING

ALLIGATION: SYSTEM OF EQUATIONS METHOD

EJ has an older prescription in the system to be dispensed as needed for athlete's foot. The prescription is for 68% of terbinafine cream in a corticosteroid base. You have 92% terbinafine cream as stock formulation, with only 14g left, as the product seems to be on back-order. You have more than enough corticosteroid base in stock. What is the maximum quantity (mass) of 68% terbinafine cream that you can compound for EJ right now?



We know that 74% of the mixture is composed of 92% terbinafine cream, of which we have only 14g. We would like to know how much cream in total we can compound.

Cross-multiply: $\frac{14 \text{ g}}{74\%} = \frac{x}{100\%}$ $x = 18.9 \text{ g}$

Therefore, we can make a maximum quantity of **18.9g of 68% terbinafine cream**

COMPOUNDING

* Determining the minimum weighable quantity (MWQ) of a balance

In practice, you may be asked to compound or weigh out certain medications

- The lowest weight that can be accurately measured is the **Minimum Weighable Quantity (MWQ)**
- The maximum acceptable error is usually 5%
- Often, you are at the maximum acceptable error (5%) when weighing the MWQ which is called the **sensitivity**

$$\text{Sensitivity} = \text{Maximum Acceptable Error (5\%)} \times \text{MWQ}$$

Example

Your lab has recently purchased a new balance with a sensitivity of 0.9 mg. Calculate the minimum weighable quantity (MWQ) of the balance.

$$\text{MWQ} = \frac{\text{sensitivity}}{\text{maximum error of 5\%}} = \frac{0.9\text{mg}}{0.05} = 18\text{mg}$$

The minimum amount you can reliably weigh using this balance is 18mg. By “reliably” we mean that whatever mass the scale reports is guaranteed to have at most 5% error.

COMPOUNDING

ALIQOT DILUTION

You are asked to compound a cream that requires 7 mg of drug Y, but the MWQ is 100 mg. How will you compound this cream?

$$\frac{\text{(A) Amount of drug in dilution}}{\text{(B) dilution volume}} = \frac{\text{(C) Amount of drug in aliquot}}{\text{(D) aliquot volume}}$$



Note: An aliquot is a portion of a larger whole. If we had 5mg of drug into a total mixture of 25mg, we have an aliquot which is 1/5 the total.

COMPOUNDING

ALIQOT DILUTION

You are asked to compound a cream that requires 7 mg of drug Y, but the MWQ is 100 mg. How will you compound this cream?

$$\frac{\text{(A) Amount of drug in dilution}}{\text{(B) dilution volume}} = \frac{\text{(C) Amount of drug in aliquot}}{\text{(D) aliquot volume}}$$

1. Decide on a dilution factor:

$$\frac{\text{MWQ}}{C} = \text{minimum dilution factor needed} \rightarrow \frac{100 \text{ mg}}{7 \text{ mg}} = 14.3 = 15$$

2. Calculate A by multiplying C times dilution factor:

$$\begin{aligned} A &= C \times \text{dilution factor} \\ &= 7 \text{ mg} \times 15 \\ &= 105 \text{ mg} \end{aligned}$$

3. Calculate B by multiplying A times dilution factor:

$$\begin{aligned} B &= A \times \text{dilution factor} \\ &= 105 \text{ mg} \times 15 \\ &= 1575 \text{ mg} \end{aligned}$$

In practice, you will need to mix 105 mg of drug Y with 1470 mg of diluent [(B) – (A) = diluent weight] to form this 1575 mg dilution.

4. Solve for D:

$$D = \frac{B * C}{A} = \left(\frac{1575 \text{ mg} * 7 \text{ mg}}{105 \text{ mg}} \right)$$

In practice, you will weigh 105 mg of the dilution and use this aliquot to compound the prescribed cream

COMPOUNDING

SUPPOSITORIES

When compounding suppositories, you must know how much base and how much active ingredient to mix to obtain the fixed volume of the suppository mold.

Example A-1

You would like to compound some acetaminophen suppositories for a dog with pain. A pure lanolin suppository has a mass of 1 g. The same size suppository containing 100 mg of acetaminophen has a mass of 1.05 g.

What mass of lanolin do you need to make 14 suppositories of 100 mg acetaminophen the dog?

Example A-2

What is the mass of lanolin displaced in each suppository?

Example A-3

What is the density factor of acetaminophen?

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SUPPOSITORIES

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What mass of lanolin do you need to make 14 suppositories of 100 mg acetaminophen the dog?

Mass of pure lanolin in a single pure lanolin suppository =
Mass of a single combination suppository =

Mass of acetaminophen in a single combination suppository =
Mass of lanolin in a single combination suppository =

Therefore mass of lanolin needed in 14 suppositories =

COMPOUNDING

SUPPOSITORIES

When compounding suppositories, you must know how much base and how much active ingredient to mix to obtain the fixed volume of the suppository mold.

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What mass of lanolin do you need to make 14 suppositories of 100 mg acetaminophen the dog?

Mass of pure lanolin in a single pure lanolin suppository = 1 g

Mass of a single combination suppository = 1.05 g

Mass of acetaminophen in a single combination suppository = 100 mg = 0.1 g

Mass of lanolin in a single combination suppository = $1.05 \text{ g} - 0.1 \text{ g} = 0.95 \text{ g}$

Therefore mass of lanolin needed in 14 suppositories = $14 \times 0.95 \text{ g} = 13.3 \text{ g of lanolin}$

COMPOUNDING

SUPPOSITORIES

When compounding suppositories, you must know how much base and how much active ingredient to mix to obtain the fixed volume of the suppository mold.

Example A-2

You would like to compound some acetaminophen suppositories for a dog with pain. A pure lanolin suppository has a mass of 1 g. The same size suppository containing 100 mg of acetaminophen has a mass of 1.05 g.

What is the mass of lanolin displaced in each suppository?

Mass of lanolin displaced = Lanolin in pure supp. – Lanolin in compounded supp.

COMPOUNDING

SUPPOSITORIES

When compounding suppositories, you must know how much base and how much active ingredient to mix to obtain the fixed volume of the suppository mold.

Example A-2

You would like to compound some acetaminophen suppositories for a dog with pain. A pure lanolin suppository has a mass of 1 g. The same size suppository containing 100 mg of acetaminophen has a mass of 1.05 g.

What is the mass of lanolin displaced in each suppository?

Mass of pure lanolin in a single pure lanolin suppository = 1 g

Mass of lanolin displaced = Lanolin in pure supp. – Lanolin in compounded supp.
= 1 g – 0.95 g
= 0.05 g

COMPOUNDING

SUPPOSITORIES

When compounding suppositories, you must know how much base and how much active ingredient to mix to obtain the fixed volume of the suppository mold.

Example A-3

You would like to compound some acetaminophen suppositories for a dog with pain. A pure lanolin suppository has a mass of 1 g. The same size suppository containing 100 mg of acetaminophen has a mass of 1.05 g.

What is the density factor of acetaminophen?

$$\text{density factor} = \frac{\text{mass of acetaminophen}}{\text{mass of lanolin displaced}}$$

COMPOUNDING

SUPPOSITORIES

When compounding suppositories, you must know how much base and how much active ingredient to mix to obtain the fixed volume of the suppository mold.

Example A-3

You would like to compound some acetaminophen suppositories for a dog with pain. A pure lanolin suppository has a mass of 1 g. The same size suppository containing 100 mg of acetaminophen has a mass of 1.05 g.

What is the density factor of acetaminophen?

$$\begin{aligned}\text{density factor} &= \frac{\text{mass of acetaminophen}}{\text{mass of lanolin displaced}} \\ &= \frac{0.1 \text{ g}}{0.05 \text{ g}} \\ &= 2\end{aligned}$$

REVIEW QUESTION – DOSE CALCULATION

A child weighing 27 lbs requires the combination antibiotic sulfamethoxazole and trimethoprim for the treatment of a bacterial infection. The treatment dose is: 8-12 mg trimethoprim/kg/day PO divided q12h. The physician tells you that she wants to administer 12 mg of the trimethoprim component/kg/day po divided twice daily. This antibiotic is commercially available as an oral suspension that contains sulfamethoxazole 40 mg/mL and trimethoprim 8 mg/mL. Please calculate the daily dose in mL for this patient.

- a) 8 mL po BID of the suspension
- b) 9 mL po BID of the suspension
- c) 7 mL po BID of the suspension
- d) 6 mL po BID of the suspension

REVIEW QUESTION – ALLIGATION

CROSS-SUBTRACT METHOD

You need to prepare 1,000 mL of a 30% sodium chloride (NaCl) solution. Available stock solutions: 70% sodium chloride, 50% sodium chloride, 20% sodium chloride. You need to determine how much of each stock solution should be mixed to create the required 1000mL of 30% sodium chloride.

REVIEW QUESTION – ALLIGATION

CROSS-SUBTRACT METHOD

You need to prepare 1,000 mL of a 30% sodium chloride (NaCl) solution. Available stock solutions: 70% sodium chloride, 50% sodium chloride, 20% sodium chloride. You need to determine how much of each stock solution should be mixed to create the required 1000mL of 30% sodium chloride.

70%		10 parts
50%		10 parts
	30%	
20%		20 + 40 = 60 parts

$10/80 * 1000 \text{ mL} = 125 \text{ mL of 70\%}$

$10/80 * 1000 \text{ mL} = 125 \text{ mL of 50\%}$

$60/80 * 1000\text{mL} = 750 \text{ mL of 20\%}$

Total = 80 parts

CHEMISTRY

Qualifying Exam Preparatory Course

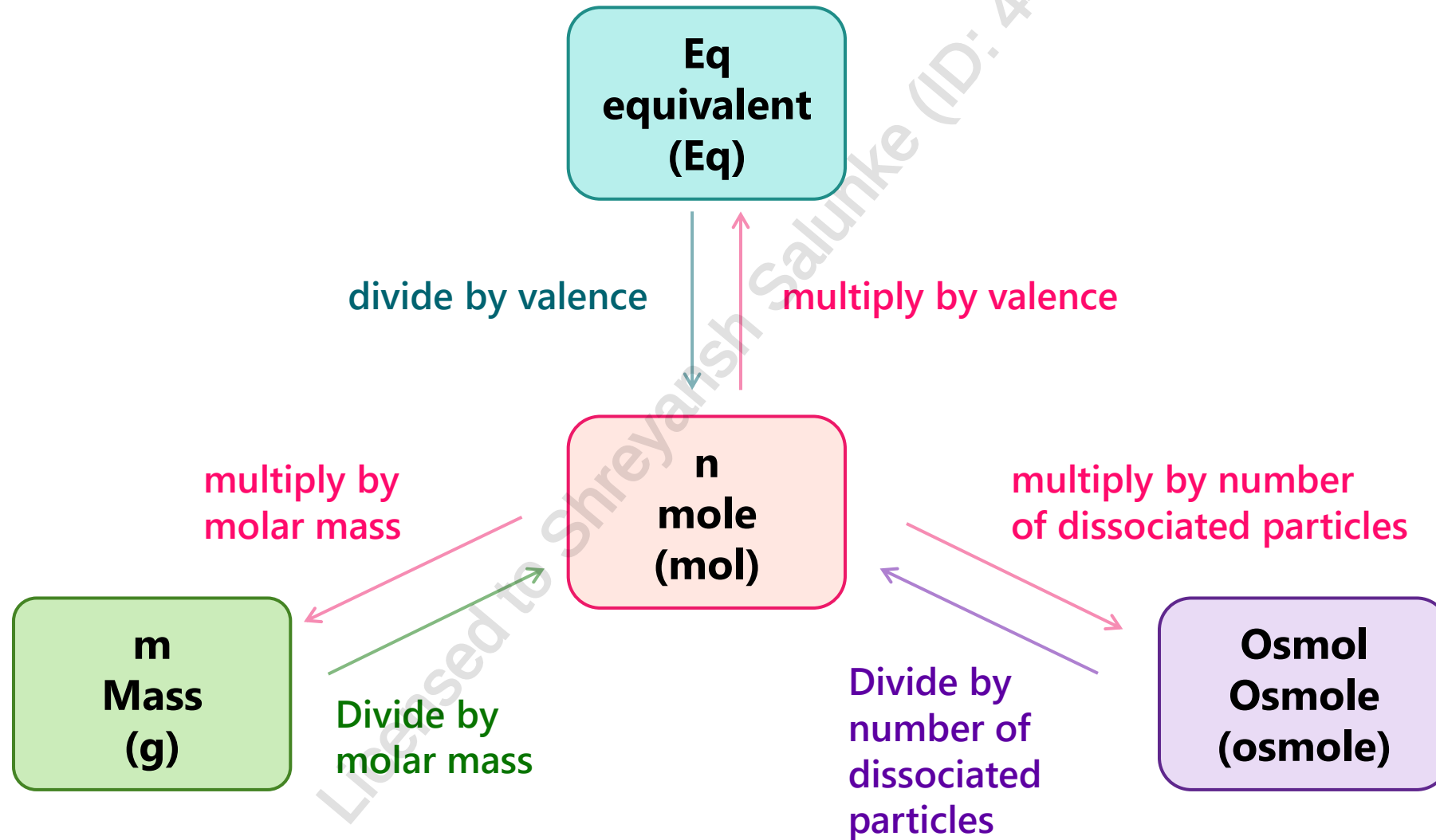


Pharm*Achieve*

CHEMISTRY DEFINITIONS

Term	Definition
Solute	minor component or dissolved component of a solution
Solvent	major component or dissolving component of a solution
Molarity	amount of substance/solute in a volume of solution, expressed in mol/L
Osmolarity	concentration of solute particles in a solution per L
Molality	measure of the number of moles of solute in a solution corresponding to 1kg or 1000g of solvent
Osmolality	concentration of solute particles in 1 kg of solvent
Valence	amount of bonds that an element can make
Equivalence	measure of the chemical activity of amount of charge of a substance

MOLES, EQUIVALENTS, AND OSMOLES



CHEMISTRY

Milliequivalents (mEq): Chemical units used to express the concentration of electrolytes in solution

Knowing the number of milliequivalents (mEq) in ionic solutions allows clinicians to determine the total number of charges present and if the charges are imbalanced, we can add the appropriate proportions of electrolytes needed to achieve electrical neutrality

$$\text{mEq} = \frac{\text{mass in mg} \times \text{valence of ion}}{\text{molecular mass in g/mol}}$$

Example

An over-the-counter antacid formulation contains 5mg of $\text{Al}(\text{OH})_3$ and 10mg of CaCO_3 (aluminum hydroxide and calcium carbonate). Calculate the milliequivalents of Al^{3+} and Ca^{2+} in this formulation. The valence of Ca is 2 and the valence of Al is 3. The molecular mass of $\text{Al}(\text{OH})_3$ is 78g/mol and the molecular mass of CaCO_3 is 100g/mol.

CHEMISTRY

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For $\text{Al}(\text{OH})_3$:

Molecular mass = 78 g/mol

$$\text{mEq} = \frac{5 \text{ mg} \times 3}{78 \text{ g/mol}} = 0.19 \text{ mEq}$$

For CaCO_3

Molecular mass = 100 g/mol

$$\text{mEq} = \frac{10 \text{ mg} \times 2}{100 \text{ g/mol}} = 0.2 \text{ mEq}$$

CHEMISTRY

MILLIEQUIVALENTS (MEQ): CALCULATING ELEMENTAL DOSES

Example

KR has a history of anemia and she takes 10mL BID of an iron supplement suspension containing 5 mEq/L of Fe^{3+} . The molecular mass of Fe is 56g/mol. What mass in mg of Fe^{3+} does KR receive in a daily dose of her supplement?

1. KR receives a daily dose of:

2. Find equivalent mg for 1 mEq of Fe^{3+} :

$$\text{mEq} = \frac{\text{mass in mg of an ion} \times \text{valence of ion}}{\text{molecular mass in g/mol}}$$

3. Mass of Fe^{3+} :

Example

After winning a hot-dog-eating contest, patient RT shows up to the pharmacy and picks up a 500mL antacid solution formulation containing 1:300 MgCO_3 . Molecular mass of MgCO_3 is 84.3g/mol. How many mEq of Mg^{2+} are there in RT's bottle?

1. Find mg of MgCO_3 in 500mL:

2. Find equivalent mEq of MgCO_3 in 500mL:

$$\text{mEq} = \frac{\text{mass in mg of an ion} \times \text{valence of ion}}{\text{molecular mass in g/mol}}$$

CHEMISTRY

MILLIEQUIVALENTS (MEQ): CALCULATING ELEMENTAL DOSES

Example

KR has a history of anemia and she takes 10mL BID of an iron supplement suspension containing 5 mEq/L of Fe^{3+} . The molecular mass of Fe is 56g/mol. What mass in mg of Fe^{3+} does KR receive in a daily dose of her supplement?

1. KR receives a daily dose of: $5 \frac{\text{mEq}}{\text{L}} \times 0.02 \text{ L} = 0.1 \text{ mEq}$

2. Find equivalent mg for 1 mEq of Fe^{3+} :
$$\text{mEq} = \frac{\text{mass in mg of an ion} \times \text{valence of ion}}{\text{molecular mass in g/mol}}$$

$$1 \text{ mEq} = \frac{(\text{mass in mg of Fe } 3+) \times (3)}{56 \text{ g/mol}}$$

$$\text{mass of Fe } 3+ = 18.7 \text{ mg} \quad 1 \text{ mEq of Fe } 3+ = 18.7 \text{ mg}$$

3. Mass of Fe^{3+} : $1 \text{ mEq Fe } 3+ = 18.7 \text{ mg}$

$$\frac{1 \text{ mEq of Fe } 3+}{18.7 \text{ mg}} = \frac{0.1 \text{ mEq of Fe } 3+}{x} \quad x = 1.87 \text{ mg}$$

Example

After winning a hot-dog-eating contest, patient RT shows up to the pharmacy and picks up a 500mL antacid solution formulation containing 1:300 MgCO_3 . Molecular mass of MgCO_3 is 84.3g/mol. How many mEq of Mg^{2+} are there in RT's bottle?

1. Find mg of MgCO_3 in 500mL:

$$\frac{1 \text{ g}}{300 \text{ mL}} = \frac{x}{500 \text{ mL}} \quad x = 1.67 \text{ g} = 1670 \text{ mg}$$

2. Find equivalent mEq of MgCO_3 in 500mL:

$$\text{mEq} = \frac{\text{mass in mg of an ion} \times \text{valence of ion}}{\text{molecular mass in g/mol}}$$

$$\text{mEq of Mg } 2+ = \frac{(1670 \text{ mg}) \times (2)}{84.3 \text{ g/mol}}$$

$$\text{mEq of Mg } 2+ = 39.3 \text{ mEq}$$

CHEMISTRY

MILLIMOLES (mmol)

Millimoles (mmol): SI unit (mol) for electrolyte concentrations in solution

- A mole is a measure of the number of particles of a substance, where 1 mole is equal to 6.02×10^{23} particles.
- Equal to mEq for monovalent species
- Not just used for electrolyte solutions but rather for concentrations of all solutions

Example

How many millimoles of sodium phosphate (MW: 138 g/mol) are present in 100g of the substance?

$$\frac{100 \text{ g}}{138 \text{ g/mol}} = 0.725 \text{ mol} = 725 \text{ mmol}$$

CHEMISTRY

MILLIOSMOLES (mOsmols):

Milliosmoles (mOsmols): number of millimoles of a particular substance multiplied by the number of ions or parts that this particular substance can dissociate into when placed in a solution

- Knowing the number of mOsmol in a solution allows you to classify a solution's osmolarity as **hypotonic, hypertonic, or isotonic**.

Example

How many mOsmol are there in 50mL of a 3% w/v $\text{Ca}_3(\text{PO}_4)_2$ solution that fully dissociates into its ions? Molar mass of $\text{Ca}_3(\text{PO}_4)_2$ is 310g/mol

$\text{Ca}_3(\text{PO}_4)_2$ dissociates into 5 ions: 3 x Ca^{2+} and 2 x PO_4^{3-}

Molecular mass of $\text{Ca}_3(\text{PO}_4)_2 = 310 \frac{\text{g}}{\text{mol}} = 310 \frac{\text{mg}}{\text{mmol}}$

Therefore, 1 mmol of $\text{Ca}_3(\text{PO}_4)_2 = 310 \text{ mg} = 5 \text{ mOsmol}$

The mass of $\text{Ca}_3(\text{PO}_4)_2$ in the specified solution: $50\text{mL} \times \frac{3\text{g}}{100\text{mL}} = 1500 \text{ mg}$

Cross-multiplication:

$$\frac{1500 \text{ mg}}{x} = \frac{310 \text{ mg}}{5 \text{ mOsmol}}$$

$$x = 24.2 \text{ mOsmol}$$

CHEMISTRY

Osmolarity (mOsmol/L): The measure of osmotic concentration in a solution

- Osmotic pressure is proportional to the total number of particles present in solution
- For nonelectrolytes, molarity and osmolarity are equal
- Species that dissociate in solution have greater osmolarity than those that do not

$$\frac{\text{mOsmol}}{\text{L}} = \frac{\text{concentration } \left(\frac{\text{mg}}{\text{L}}\right)}{\text{Molecular weight } \left(\frac{\text{mg}}{\text{mmol}}\right)} \times \text{number of species}$$

Example

What is the osmolarity of a 1000 mL solution containing 50g of dextrose, 4.5g of sodium chloride and 1.5g of potassium chloride? Molecular mass of dextrose is 180g/mol, of NaCl is 58.5g/mol, and of KCl is 74.5g/mol.

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What is the osmolarity of a 1000 mL solution containing 50g of dextrose, 4.5g of sodium chloride and 1.5g of potassium chloride? Molecular mass of dextrose is 180g/mol, of NaCl is 58.5g/mol, and of KCl is 74.5g/mol.

$$\text{Dextrose: } \frac{50000 \text{ mg/L}}{180 \text{ mg/mmol}} * 1 = 278 \text{ mOsmol/L}$$

$$\text{NaCl: } \frac{4500 \text{ mg/L}}{58.5 \text{ mg/mmol}} * 2 = 154 \text{ mOsmol/L}$$

$$\text{KCl: } \frac{1500 \text{ mg/L}}{74.5 \text{ mg/mmol}} * 2 = 40 \text{ mOsmol/L}$$

$$\text{Total} = 472 \text{ mOsmol/L}$$

BUFFER SOLUTIONS

- Resist changes in pH to maintain ion activity within narrow limits
- Useful to make injections and ophthalmic solutions, as well as to help maintain product stability
- Often made of a weak acid and a salt of the acid OR weak base and salt of the base
- HA refers to the acid. A⁻ refers to the salt of acid $\text{HA} + \text{H}_2\text{O} \leftrightarrow \text{H}_3\text{O}^+ + \text{A}^-$
- B refers to the base. HB⁺ refers to the salt of base. $\text{B} + \text{H}_2\text{O} \leftrightarrow \text{HB}^+ + \text{OH}^-$

Dissociation Constant: Strength of an acid or base

- K_a = dissociation constant of an acid
- K_b = dissociation constant of a base

$$K_a = \frac{[\text{H}^+][\text{A}^-]}{[\text{HA}]} \quad K_b = \frac{[\text{HB}^+][\text{OH}^-]}{[\text{B}]}$$

- $\text{p}K_a = -\log(K_a)$
- $\text{p}K_b = -\log(K_b)$
- $\text{pH} + \text{pOH} = \text{p}K_w = 14$
- $\text{pH} = -\log[\text{H}^+]$

Henderson-Hasselbalch Equation

$$\text{pH} = \text{p}K_a - \log \frac{[\text{HA}]}{[\text{A}^-]}$$

$$\text{pOH} = \text{p}K_b + \log \frac{[\text{HB}^+]}{[\text{B}]}$$

$$\text{pH} = 14 - \text{p}K_b + \log \frac{[\text{B}]}{[\text{HB}^+]}$$

PH OF SOLUTIONS

HENDERSON-HASSELBALCH (HH) EQUATION

$$\text{pH} = \text{pK}_a - \log \frac{[\text{HA}]}{[\text{A}^-]}$$

$$\text{pH} = -\log[\text{H}^+]$$

$$\text{pOH} = \text{pK}_b + \log \frac{[\text{HB}^+]}{[\text{B}]}$$

Example

A solution of a certain amino acid has a pKa of 3.8. The ratio of ionized to un-ionized amino acid in the solution is 120:1. What is the pH of this solution? HA refers to the acid (*un-ionized*), A⁻ refers to the base (*ionized*)

Example

A solution of magnesium carbonate has a carbonate ion concentration of 5M and a bicarbonate concentration of 2M. The K_b of carbonate is 2.1 x 10⁻⁴. Assuming the magnesium ions do not contribute to the acidity or basicity, what is this solution's pH?

PH OF SOLUTIONS

HENDERSON-HASSELBALCH (HH) EQUATION

$$\text{pH} = \text{pK}_a - \log \frac{[\text{HA}]}{[\text{A}^-]}$$

$$\text{pH} = -\log[\text{H}^+]$$

$$\text{pOH} = \text{pK}_b + \log \frac{[\text{HB}^+]}{[\text{B}]}$$

Example

A solution of a certain amino acid has a pKa of 3.8. The ratio of ionized to un-ionized amino acid in the solution is 120:1. What is the pH of this solution? HA refers to the acid (*un-ionized*), A⁻ refers to the base (*ionized*)

Ratio of HA to A⁻: 1/120 = 0.0083

$$\begin{aligned}\text{pH} &= \text{pK}_a - \log \frac{[\text{HA}]}{[\text{A}^-]} \\ &= 3.8 - \log 0.0083 \\ &= 3.8 - (-2.1) \\ &= 5.9\end{aligned}$$

Example

A solution of magnesium carbonate has a carbonate ion concentration of 5M and a bicarbonate concentration of 2M. The K_b of carbonate is 2.1 x 10⁻⁴. Assuming the magnesium ions do not contribute to the acidity or basicity, what is this solution's pH?

$$\text{pK}_b = -\log K_b = -\log 2.1 \times 10^{-4} = 3.68$$

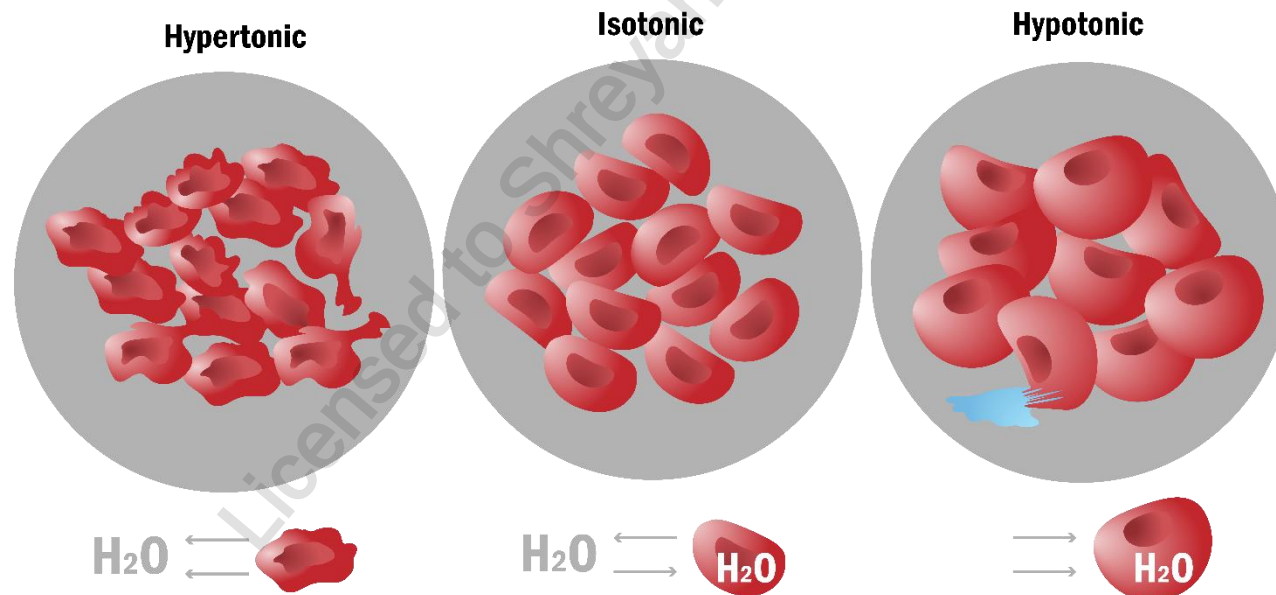
$$\frac{[\text{HB}^+]}{[\text{B}]} = \frac{2 \text{ M}}{5 \text{ M}}$$

$$\text{pOH} = \text{pK}_b + \log \frac{[\text{HB}^+]}{[\text{B}]} = 3.68 + \log 0.4 = 3.28$$

$$\text{pH} = 14 - \text{pOH} = 14 - 3.28 = 10.72$$

ISOTONICITY CALCULATIONS AND FREEZING POINT DATA

- Why are isotonicity calculations important?
 - To prepare parenteral and ophthalmic solutions, they must be isotonic with bodily fluids to prevent cell damage and maximize efficacy
- How can we make solutions isotonic? Freezing point depression data(ΔT_f)
 - $\Delta T_{f \text{ bodily fluid}} = -0.52^\circ\text{C}$
 - $\Delta T_{f \text{ bodily fluid}} = \Delta T_{f \text{ drug solution}} = \text{isotonic}$



ISOTONICITY CALCULATIONS AND FREEZING POINT DATA

Example

How many mg of sodium chloride and atropine sulphate are needed to prepare 45 mL of a 1% atropine sulphate isotonic ophthalmic solution? (Given: $\Delta T_f^{1\%} \text{ atropine sulphate} = -0.07^\circ\text{C}$, $\Delta T_f^{1\%} \text{ sodium chloride} = -0.58^\circ\text{C}$, $\Delta T_{f \text{ bodily fluid}} = -0.52^\circ\text{C}$)

ISOTONICITY CALCULATIONS AND FREEZING POINT DATA

Example

How many mg of sodium chloride and atropine sulphate are needed to prepare 45 mL of a 1% atropine sulphate isotonic ophthalmic solution?

1. Calculate by how much the freezing point needs to be lowered by:

$$\Delta T_f^{1\%} \text{ atropine sulphate} = -0.07^\circ\text{C}$$

$$\Delta T_f^{1\%} \text{ sodium chloride} = -0.58^\circ\text{C}$$

$$\Delta T_f \text{ lacrimal fluid} = -0.52^\circ\text{C}$$

$$0.52^\circ - 0.07^\circ = 0.45^\circ$$

Thus, the freezing point must be lowered an additional 0.45°

2. Calculate the amount of sodium chloride required to lower the freezing point:

$$\frac{(1\% \text{ sodium chloride} \times 0.45)}{0.58} = 0.78\%$$

Sodium chloride can be used to lower the freezing point of atropine sulphate

3. Calculate how much solute you need to prepare the solution:

$$45 \text{ mL} \times 1\% = 0.45 \text{ g} = \mathbf{450 \text{ mg atropine sulphate}}$$

$$45 \text{ mL} \times 0.78\% = 0.351 \text{ g} = \mathbf{351 \text{ mg sodium chloride}}$$

To prepare 45 mL of solution

PARENTERAL MEDICATIONS

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PARENTERAL MEDICATIONS

- Medications not administered by mouth or through the GI tract
 - Intramuscular – IM
 - Subcutaneous – Sub-Q
 - Intradermal – ID
 - Intravenous - IV
 - Transdermal
- Some medications lose potency quickly in solution and are supplied in a powdered form to be reconstituted using the appropriate diluent
 - Instructions are provided with the medication and different solvents can be used (sterile water, saline, bacteriostatic solution)
 - After reconstitution, medication has a very limited shelf life

AMPICILLIN

For I.V. use
500mg

CAUTION: Federal law prohibits
dispensing without prescription.

Add 1.7mL of sterile water for a final concentration
of 250mg/mL
Stable for 1 hour.
Read accompanying circular.

EXPIRES 09/09/09

ERYTHROMYCIN_F

or I.V. use
250mg

CAUTION: Federal law prohibits
dispensing without prescription.

Add 19.2mL of sterile water for a final concentration
of 250mg/20mL
Stable for 48 hours at room temperature or 7 days
in the refrigerator.
Read accompanying circular.

EXPIRES 09/09/09

PARENTERAL MEDICATIONS

IV Infusion

Sterile aqueous preparations administered into the vein in relatively large volumes

- Used to extend blood volume, provide electrolytes, nutrients or medication
- Widely used in emergency care units and surgeries
- **Continuous infusions:** run large volumes of fluid into a vein uninterrupted
- **Intermittent infusions:** are administered at scheduled times
- **IV push:** rapid infusion of medication

Infusion flow rate

- Drops per minute or amount of drug per time or approximate amount of time for total volume

$$\text{FlowRate(drops/min)} = \frac{\text{Volume (mL)} * \text{Drip set } (\frac{\text{drops}}{\text{mL}})}{\text{Time (minutes)}}$$

Example

A medication order calls for 1000mL to be administered over an 8-hour period. Using an IV administrator set that delivers 10drops/mL, how many drops per minute should be delivered to the patient?

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Example

A medication order calls for 1000mL to be administered over an 8-hour period. Using an IV administrator set that delivers 10drops/mL, how many drops per minute should be delivered to the patient?

$$\text{Flow rate} = \frac{1000\text{mL} * 10 \frac{\text{drops}}{\text{mL}}}{8\text{hrs} * 60 \frac{\text{min}}{\text{hr}}}$$

$$\text{Flow rate} = \frac{10\,000\text{drops}}{480\text{min}}$$

$$\text{Flow rate} = 20.8 \frac{\text{drops}}{\text{min}}$$

PRACTICE QUESTION

You prepare a potassium chloride (KCl) solution containing Dextrose 5% and KCl 20 mmol/L. If your patient is receiving this intravenous solution as a continuous infusion at the rate of 50 mL/hr, how many millimoles of potassium chloride will your patient have received in 12 hours?

- a) 12 mmol/hr
- b) 12 mmol
- c) 12,000 mmol
- d) 1 mmol

TOTAL PARENTERAL NUTRITION (TPN)

Qualifying Exam Preparatory Course



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PARENTERAL MEDICATIONS

TOTAL PARENTERAL NUTRITION

- Feeding of a patient by the intravenous infusion of fluids and basic nutrients
- Provides all the patient's nutritional requirements for patients who cannot obtain adequate nutrition by oral means
- Central administration lines placed into superior vena cava and peripheral lines into veins
 - Central lines are preferred for high [dextrose] as hypertonic solutions may damage veins
- Formulas differ based on age group and patient specific diseases
 - Includes; carbs, protein, fats, electrolytes, vitamins, trace elements and sterile water injections

Parameter	Fluid Requirement
Neonates	120-180 mL/kg per day
1-12 months	150 mL/kg (max 200mL)
<10 kg	100 mL/kg per day
10-20 kg	1000mL + 50 ml/kg over 10kg
>20 kg	1500mL + 20 mL/kg over 20kg
Body Surface Area	1500 mL/m ² per day
Caloric Requirement	1.2 mL/kcal per day

PARENTERAL MEDICATIONS

TPN: CALORIC REQUIREMENTS

Carbohydrate Requirements:

- Primary source of daily energy requirements
- Dextrose provides **3.4 kcal/gram**

Protein Requirements:

- Provided as protein hydrolysate or amino acids
- Protein solution provides **4 kcal/gram**

Fat Requirements:

- Lipids provide 30-40% of the daily calorie requirements
- 10% lipid emulsion provides **11 kcal/gram**

Fiber Requirements:

- **14 g** of fiber for each 1000 calories consumed generally recommended

Patient type	Amount of protein/day
Unstressed	0.8 g/kg
Mildly stressed	0.8-1 g/kg
Moderately stressed	1.1-1.5 g/kg
Severely stressed	1.5-2 g/kg
Renal Dialysis	1.2 g/kg
Severely Burned	3 g/kg

PARENTERAL MEDICATIONS

TPN: CALORIC REQUIREMENTS

Caloric requirements vary depending on a patient's physical state and medical condition(s)

- Harris-Benedict equations are used to estimate basal energy expenditure (BEE)
- Total daily expenditure (TDE) accounts for activity and stress factors

Males: $BEE = 66.67 + [13.75 * \text{weight(kg)}] + [5 * \text{height(cm)}] - [6.76 * \text{age(yr)}]$

Females: $BEE = 655.1 + [9.56 * \text{weight(kg)}] + [1.86 * \text{height(cm)}] - [4.68 * \text{age(yr)}]$

TDE (Male or Female) = BEE * activity factors * stress factors

Activity Factors

- Confined to bed: 1.2
- Ambulatory: 1.3

Stress Factors

- Surgery: 1.2
- Trauma: 1.3-1.5
- Infection: 1.4-1.6
- Burns: 1.5-2.1

PARENTERAL MEDICATIONS

TPN: SAMPLE PROBLEM

A 48-year-old woman is 5 ft.5 in. tall and weighs 145 lb. She is currently on TPN and 10% fat emulsion. She has no disease states or activities that would alter her nutritional requirements. Calculate the additional dextrose requirement in grams to meet her daily caloric needs. Use the Harris-Benedict equation and assume her lipid requirement is 35% of total daily calories.

Step 1

Total daily kcal required -Harris-Benedict equation:

$$\text{BEE} = 655.1 + [9.56 * \text{weight}(\text{kg})] + [1.86 * \text{height}(\text{cm})] - [4.68 * \text{age}(\text{yr})]$$

Step 2

Protein Required:

Step 3

Calculate the number of calories supplied from protein:

Step 4

Lipids required (kcal) using 35% of total daily calories:

Step 5

Carbohydrates (dextrose) required for remaining calorie requirement:

Step 6

Convert kcal of dextrose required to grams:

PARENTERAL MEDICATIONS

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Step 1

Total daily kcal required - Harris-Benedict equation:

$$\text{BEE} = 655.1 + [9.56 * \text{weight(kg)}] + [1.86 * \text{height(cm)}] - [4.68 * \text{age(yr)}]$$
$$655.1 + (9.56 * 65.9\text{kg}) + (1.86 * 165\text{cm}) - (4.68 * 48\text{yr})$$
$$= 1\,367.36 \text{ kcal}$$

Step 2

Protein Required:

$$145 \text{ lb} * \frac{1 \text{ kg}}{2.2 \text{ lb}} * \frac{0.8 \text{ g}}{1 \text{ kg}} = 52.72 \text{ g of protein}$$

Step 3

Calculate the number of calories supplied from protein:

$$52.72 \text{ g} * \frac{4 \text{ kcal}}{1 \text{ g}} = 210.88 \text{ kcal}$$

Step 4

Lipids required (kcal) using 35% of total daily calories:

$$1\,367.36 \text{ kcal} * 35\% = 478.58 \text{ kcal}$$

Step 5

Carbohydrates (dextrose) required for remaining calorie requirement:

$$1\,367.36 \text{ kcal} - 210.88 \text{ kcal} - 478.58 \text{ kcal}$$
$$= 677.9 \text{ kcal}$$

Step 6

Convert kcal of dextrose required to grams:

$$677.9 \text{ kcal} * \frac{1 \text{ g}}{3.4 \text{ kcal}} = 199.38 \text{ g}$$

PARENTERAL MEDICATIONS

TPN: COUNTING CALORIES

Example

Patient TK is admitted to the hospital for acute pancreatitis. After diagnostic tests, she is to have nothing by mouth (NPO) for 10 days and requires total parenteral nutrition. She is ordered 2L of TPN and 500 mL of lipids daily, which is composed of:

- Protein (4 Cal/g) = 1g/14mL
- Dextrose (3.4 Cal/g) = 8% w/v
- Lipids (11 Cal/g) = 10% fat emulsion

Calculate the total number of daily calories TK receives from her TPN and lipids.

Protein:

Lipids:

Dextrose:

Total Calories

PARENTERAL MEDICATIONS

TPN: COUNTING CALORIES

Example

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- Protein (4 Cal/g) = 1g/14mL
- Dextrose (3.4 Cal/g) = 8% w/v
- Lipids (11 Cal/g) = 10% fat emulsion

Calculate the total number of daily calories TK receives from her TPN and lipids.

Protein: $\frac{1 \text{ g}}{14 \text{ mL}} = \frac{x}{2000 \text{ mL}}$ $x = 142.9 \text{ g}$

Protein Calories = $142.9 \text{ g} \times 4 \text{ cal/g} = 571.6 \text{ Cal}$

Dextrose: $\frac{8 \text{ g}}{100 \text{ mL}} = \frac{x}{2000 \text{ mL}}$ $x = 160 \text{ g}$

Dextrose Calories = $160 \text{ g} \times 3.4 \text{ cal/g} = 544 \text{ Cal}$

Lipids: $\frac{10 \text{ g}}{100 \text{ mL}} = \frac{x}{500 \text{ mL}}$ $x = 50 \text{ g}$

Lipid Calories = $50 \text{ g} \times 11 \text{ cal/g} = 550 \text{ Cal}$

Total Calories = $571.6 + 544 + 550$
= **1665.6 Calories**

DOSE TITRATION CALCULATIONS

EXAMPLE:

A patient will start on prednisone 60 mg once daily (using 5mg tablets). The physician instructs him to taper the dose by 10 mg every 7 days until he reaches 20 mg daily for 1 week, then stop. How many 5 mg tablets will the pharmacist need to dispense for the entire tapering course?

- A. 272 tablets
- B. 200 tablets
- C. 370 tablets
- D. 280 tablets

DOSE TITRATION CALCULATIONS

Step 1: Work out taper schedule

- Start: 60 mg/day
- Reduce 10 mg every 7 days until 20 mg/day

So, the sequence of doses:

- Week 1: 60 mg
- Week 2: 50 mg
- Week 3: 40 mg
- Week 4: 30 mg
- Week 5: 20 mg → then stop

WEEK	DAILY DOSE	TABLETS /DAY	TOTAL TABLETS /WEEK
WEEK 1	60mg	12	84
WEEK 2	50mg	10	70
WEEK 3	40mg	8	56
WEEK 4	30mg	6	42
WEEK 5	20mg	4	28

Answer: The pharmacist must dispense **280 tablets** of 5 mg prednisone

DOSE TITRATION CALCULATIONS

EXAMPLE:

A patient has been newly prescribed sertraline 25 mg tablets for depression with a gradual titration plan. During the first week, the patient will take 25 mg once daily. In the second week, the dose increases to 50 mg once daily. By the third week, the patient will take 75 mg once daily, and from the fourth week onward the maintenance dose will be 100 mg once daily. Based on this titration schedule, the pharmacist must calculate how many 25 mg tablets are required to cover the first four weeks of therapy?

- A. 70 tablets
- B. 84 tablets
- C. 98 tablets
- D. 112 tablets

DOSE TITRATION CALCULATIONS

Step 1:

Convert each week's dose into tablets/day

- Week 1: 25 mg → 1 tab/day → 7 tabs
- Week 2: 50 mg → 2 tabs/day → 14 tabs
- Week 3: 75 mg → 3 tabs/day → 21 tabs
- Week 4: 100 mg → 4 tabs/day → 28 tabs

Step 2:

Add totals

$$7 + 14 + 21 + 28 = 70 \text{ tablets}$$

Correct Answer: **A. 70 tablets**

PRACTICE PROBLEMS

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PRACTICE PROBLEM #1

ALIQOT DILUTION

Compound a 100 mL solution containing 0.09% (w/l) acetylsalicylic acid (ASA) in water. The Minimum Weighable Quantity is 100 mg for your scale. According to Merck's Index, 1 g of ASA is soluble in 300 mL of water at 25 ° C. Outline your steps.

$$\frac{\text{(A) Amount of drug in dilution}}{\text{(B) dilution volume}} = \frac{\text{(C) Amount of drug in aliquot}}{\text{(D) aliquot volume}}$$

1. Determine weight of ASA required (C):
2. Determine solubility of ASA:
3. Decide aliquot volume (D):
 - Aliquot volume < prescription volume
 - Aliquot concentration < ASA's max solubility
4. Decide the amount of weighable drug (A):
5. Calculate the dilution volume (B):

PRACTICE PROBLEM #1

ALIQUOT DILUTION

Compound a 100 mL solution containing 0.09% (w/l) acetylsalicylic acid (MWQ is 100 mg)

$$\frac{\text{(A) Amount of drug in dilution}}{\text{(B) dilution volume}} = \frac{\text{(C) Amount of drug in aliquot}}{\text{(D) aliquot volume}}$$

1. Determine weight of ASA required (C): $0.09\% = 0.09 \text{ g} = 90 \text{ mg}$ of ASA in 100 mL

2. Determine solubility of ASA: 1 g ASA is soluble in 300 mL of water at 25°C :

$$\text{Max solubility} = \frac{1000 \text{ mg}}{300 \text{ mL}} = 3.3 \text{ mg/mL}$$



1. Dissolve 180 mg ASA in sufficient water to make 100 mL dilution.
2. Remove 50 mL aliquot from the concentrated solution. Add water to the aliquot until 100 mL solution is made

PRACTICE PROBLEM #1

ALIQOT DILUTION

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$$\frac{\text{(A) Amount of drug in dilution}}{\text{(B) dilution volume}} = \frac{\text{(C) Amount of drug in aliquot}}{\text{(D) aliquot volume}}$$

3. Decide aliquot volume (D):

- Aliquot volume < prescription volume — Try using aliquot volume of 50 mL
- Aliquot concentration < ASA's max solubility $\frac{90 \text{ mg ASA}}{50 \text{ mL}} = 1.8 \text{ mg/mL}$

4. Decide the amount of weighable drug (A):

$$\text{Dilution factor} = \frac{\text{MWQ}}{C} = \frac{100 \text{ mg}}{90 \text{ mg}} = 1.11 = 2 \rightarrow A = C \times \text{dilution factor} = 90 \text{ mg} \times 2 = 180 \text{ mg}$$

5. Calculate the dilution volume (B):

$$B = \frac{(D * A)}{C} = \frac{(50 \text{ mL} * 180 \text{ mg})}{90 \text{ mg}} = 100 \text{ mL}$$

PRACTICE PROBLEM #2

DOSAGE CALCULATIONS

While working as a community pharmacy you receive an order of 80 tablets of 60-mg phenobarbital. However, you currently only have 80-mg tablets available.

1. How many 80-mg tablets are needed to make 80 tablets of 60-mg tablets?
2. (b) If the tablets from part (a) are powdered and weight a total of 5650 mg, how many milligrams of the powder would provide the phenobarbital required?
3. (c) If the desired weight of each tablet is 120 mg, how much lactose is required as diluent?
4. (d) If the bioavailability of the 60-mg phenobarbital tablets is 0.44 compared to the 1.0 for a phenobarbital injection, how many mL of an injection containing 30 mg/mL phenobarbital would be bioequivalent to one tablet?

PRACTICE PROBLEM #2

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PRACTICE PROBLEM #2

DOSAGE CALCULATIONS

While working as a community pharmacy you receive an order of 80 tablets of 60-mg phenobarbital. However, you currently only have 80-mg tablets available.

1. How many 80-mg tablets are needed to make 80 tablets of 60-mg tablets?

$$80 \text{ tablets} \times 60\text{mg} = 4800\text{mg}$$

$$4800\text{mg} / 80\text{mg (tablets)} = 60 \text{ tablets}$$

2. (b) If the tablets from part (a) are powdered and weight a total of 5650 mg, how many milligrams of the powder would provide the phenobarbital required?

Since 60 tablets are needed exactly, then the full 5650 mg would provide the phenobarbital required.

PRACTICE PROBLEM #2

DOSAGE CALCULATIONS

While working as a community pharmacy you receive an order of 80 tablets of 60-mg phenobarbital. However, you currently only have 80-mg tablets available.

3. If the desired weight of each of the 80 tablets is 120 mg, how much lactose is required as diluent?
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PRACTICE PROBLEM #2

DOSAGE CALCULATIONS

While working as a community pharmacy you receive an order of 80 tablets of 60-mg phenobarbital. However, you currently only have 80-mg tablets available.

3. If the desired weight of each of the 80 tablets is 120 mg, how much lactose is required as diluent?

$$120 \text{ mg} \times 80 \text{ tablets} = 9600 \text{ mg total weight}$$

$$9600 \text{ mg} - 5650 \text{ mg} = \mathbf{3950 \text{ mg}}$$

4. If the bioavailability of the 60mg phenobarbital tablets is 0.44 compared to the 1.0 for a phenobarbital injection, how many mL of an injection containing 30 mg/mL phenobarbital would be bioequivalent to one tablet?

$$0.44 \times 60 \text{ mg} = 1 \times Dose_{injection}$$

$$Dose_{injection} = 26.4 \text{ mg}$$

$$Dose_{injection} \text{ in mL} = 26.4 \text{ mg} \div 30 \text{ mg/mL} = \mathbf{0.88 \text{ mL}}$$

PRACTICE PROBLEM #3

IV FLUID

The medical resident has ordered an IV solution containing dextrose 15%/NaCl 0.2% with 30mmol/L of KCl. Note that NaCl 0.2% = NaCl 34mmol/L. You have the following stock

solutions:

- NaCl 4mmol/mL
- KCl 2mmol(K⁺)/mL
- Dextrose 50%
- Sterile water for injection

Calculate the volume of sterile water needed to create a 100mL bag.

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Calculate the volume of sterile water needed to create a 100mL bag.

Dextrose 50% (50g/100mL): Need 15g of dextrose for 100mL of solution.

$15\text{g} \div 0.5\text{g/mL} = 30\text{ mL}$ of dextrose 50%

NaCl 4mmol/mL: (Note: NaCl 0.2% = NaCl 34mmol/L) Need 3.4 mmol of NaCl for 100mL of solution.

$3.4\text{ mmol} \div 4\text{ mmol/mL} = 0.85\text{ mL}$ of NaCl 4mmol/mL

KCl 2mmol/mL: Need 3 mmol of KCl for 100mL of solution.

$3\text{mmol} \div 2\text{ mmol/mL} = 1.5\text{mL}$ of KCl 2mmol/mL

Sterile water for injection: Volume needed = 100mL – total volume of all above components

$100\text{mL} - 30\text{mL} - 0.85\text{mL} - 1.5\text{mL} = 67.65\text{mL}$ of sterile water for injection

CHANGE LOG

July 2025

- Content reviewed; minor formatting changes throughout
- Slide 22: Changed patch dose to mcg per hour, added more values and a note about dose reduction not required
- Slides 23-25: Added practice questions
- Slides 67-68: Added units for molecular mass (g/mol)
- Slide 76: Changed magnesium carbonate ion to bicarbonate
- Slide 79: Added givens
- Slide 85: Added practice question

September 2025

- Slide 19: Removed parenteral column and tramadol equivalence dose since morphine equivalence is not reliably established
- Slides 82 and 83: Changed from 0.9% NaCl to 1% NaCl
- Slide 97-100: New dose titration calculations

October 2025

- Slides 32 and 33: Removed 'ideal' from the Cockcroft-Gault equation
- Slide 34: Used actual body weight instead of ideal body weight and updated answer

January 2026

- Slide 26: Added a slide with an example on converting PO opioid doses to fentanyl
- Slide 33: Added slide on how to choose which weight to use
- Slide 93: Edited protein requirement to 4 kcal/g
- Slides 96, 97 and 98: Adjusted numbers based on updated protein requirement